

Final Project Selection Guide

DSE6066: Applied Business Forecasting

Project Overview

Instead of a generic forecasting project, you will select one of five specialized **Industry Tracks**. Each track represents a classic “high-stakes” forecasting problem encountered in the professional world.

Your goal is to build a robust forecasting pipeline, ranging from EDA to advanced modeling, and communicate your findings in a professional memo.

The 5 Industry Tracks

1. Energy: Grid Load Demand

- **The Problem:** Predict hourly electricity consumption to help utility companies maintain grid stability and manage costs.
- **Detailed Guide:** [Track 1 Case Study](#)
- **Data Pointer:** [PJM Interconnection Hourly Load Data \(Kaggle\)](#)

2. Finance: Returns & Risk (M6 Competition)

- **The Problem:** Forecast monthly stock returns and volatility for a basket of assets.
- **Detailed Guide:** [Track 2 Case Study](#)
- **Data Pointer:** [M6 Financial Forecasting Competition Data](#)

3. Supply Chain: Intermittent Demand (M5 Competition)

- **The Problem:** Forecast daily sales for Walmart products across different departments and stores.
- **Detailed Guide:** [Track 3 Case Study](#)
- **Data Pointer:** [M5 Forecasting - Accuracy \(Kaggle\)](#)

4. Retail/Marketing: Promotion Analysis

- **The Problem:** Forecast daily sales for European drug stores and quantify the ROI of promotions.
- **Detailed Guide:** [Track 4 Case Study](#)
- **Data Pointer:** [Rossmann Store Sales \(Kaggle\)](#)

5. Economics: Macro-Indicator Forecasting

- **The Problem:** Forecast the next 12 months of US Inflation (CPI) or Unemployment for a policy briefing.
 - **Detailed Guide:** [Track 5 Case Study](#)
 - **Data Pointer:** [FRED \(Federal Reserve Economic Data\)](#)
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Deliverables (Common to all Tracks)

Regardless of the track you choose, your submission must include: 1. **Professional Memo (PDF)**: A executive-level summary of your results. 2. **Reproducible Code**: A `.qmd` or `.ipynb` file that executes the entire pipeline. 3. **Visualization**: Clear, branded charts showing your data, decomposition, and final forecasts.

Pro Tip: Choose the track that aligns with your career interests. If you want to work in Tech or E-commerce, the **M5 (Supply Chain)** track is highly valued by employers like Amazon and Walmart.